



EDUCATION

The Hong Kong University of Science and Technology B.Eng. Sep. 2023 - Jun. 2027 (Expected)

• Integrative Systems & Design (Minor in Robotics) · CGA 3.43/4.3 · MCGA 3.52/4.3 · Rank 4/28

INTERNSHIP EXPERIENCE

Mondo Tech Robotics Algorithm Intern May 2026 - Present

- **Robot Data Infrastructure:** Built the full Sonic teleoperation and data pipeline for an in-house humanoid and Unitree G1 + dexterous hands (capture, transfer, storage), powering a large-scale data-collection factory.
- **Data Standard & Platform:** Helped define the unified Mondo training-data format for multi-embodiment data; built a data-management platform with visualization, versioning and data-mix configuration.
- **World-Model Training & Evaluation:** Ran controlled comparisons across video-generative backbones (Cosmos, Wan2.2, GR00T) and data mixes, using open-loop and closed-loop G1 real-robot evaluation to guide selection.

RESEARCH EXPERIENCE

DexBotRL Real-World RL for Whole-Body Dexterous Manipulation Jun. 2026 - Present

- Built a real-world RL system for whole-body dexterous manipulation on Unitree G1 + BrainCo Revo2 tactile hands (GR00T, RLinf); Actor-Critic improved success and robustness under limited real data, +22% success in simulation.
- Addressed sparse rewards, high human intervention and single-task generalization by exploring agent-driven dense rewards / multi-head critics, low-intervention online learning and a multi-task shared Actor-Critic.

Memory-Augmented VLA Undergraduate Research Program Feb. 2026 - Jul. 2026

- Designed Memory VLA to mitigate memory forgetting in long-horizon manipulation, combining hierarchical task decomposition with a fixed-capacity compressive memory that preserves key history for long-range decisions.
- Led the VLA architecture design and iteration, including core components (subtask boundary detector, VAE compression); deployed and benchmarked on RMBench and VLABench, with ongoing memory-VLA literature review.

4D World Model for Robotic Manipulation ICML 2026 (Accepted) Jan. 2026 - Feb. 2026

- Built an embodied 4D world model generating multi-view-consistent 3D/4D scene representations from single-view RGBD input, enabling an imagine-then-act paradigm for complex manipulation.
- Set up the real-robot platform (Piper arm + multi-view RGBD array + compute) and acquisition pipeline (multi-camera calibration, time sync, cross-view alignment); contributed to the 4D manipulation dataset.

PROJECT EXPERIENCE

Indoor Structural Semantic Segmentation Personal Research Project Dec. 2025 - Jan. 2026

- Designed a point-cloud semantic segmentation system for indoor structures, finely identifying ceilings, walls and other architectural elements, reaching 95% multi-class segmentation accuracy in real-world scenes.
- Built on Sonata pretraining + PTV3 backbone; owned annotation, dataset construction and training end-to-end.

HKUST RoboMaster Team ENTERPRIZE Head of Mechanical Dept. Sep. 2023 - Aug. 2025

- Designed core mechanics (serial-leg infantry — Best Infantry 2025, UAV, gimbal), all open-sourced.
- Led department management and cross-team coordination; built the design workflow and documentation system.

Personal Projects

- Bipedal RL Walking (IsaacGym) • Omnidirectional Wheelchair • 6-DoF Underwater Robot
- 3D-Printer Auto Material System (AMS) • RFID Smart Tool Cabinet • Desktop CNC Router

AWARDS

Scholarship: HKSAR Government Talent Development Scholarship (2024, 2025)

Competition: Champion — RoboMaster 2025 Univ. League & 2024 RMUC International

SKILLS

Embodied AI: VLA, world models, GR00T training/deployment; extensive real-robot and robotics R&D experience.

Engineering: Python, Linux, Git, full-stack embedded-to-algorithm; strong mechanical design (SolidWorks).